

Chapter 5: Firewall Concepts and Configuration

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Duration: 90 minutes

Objective:

Understand the **fundamentals of firewalls** and **develop practical skills** in configuring packet filters, stateful inspection, NAT, port forwarding, and ACL rules.

Lecture Outline

1. Introduction to Firewalls
2. Firewall Fundamentals
3. Packet vs. Stateful Inspection
4. Network Address Translation (NAT)
5. Port Forwarding
6. Access Control Lists (ACLs)
7. Configuration Demonstration
8. Troubleshooting & Best Practices

1. Introduction to Firewalls

Definition:

A firewall is a network security device that monitors and controls incoming and outgoing traffic based on predetermined security rules.

Purpose:

- Enforce network security policies
- Block unauthorized access
- Separate trusted and untrusted zones
- Record and analyze traffic events

Firewall in Network Security Architecture

```
[Internet]
↓
[Perimeter Firewall]
↓
[DMZ: Web/Email Servers]
↓
[Internal Network]
↓
[Host-Based Firewalls]
```

Defense-in-Depth Role:

Firewalls provide **first-line traffic control**, complemented by IDS/IPS and endpoint security.

2. Firewall Fundamentals

Concept	Description	Example
Rule	Defines which packets are allowed or denied	"Allow TCP 80 from LAN to Internet"
Policy	Collection of rules applied to interfaces	"Inbound Policy: Deny All Except HTTPS"
Zone	Logical network segment grouping	Internal, DMZ, External
Default Action	Behavior when no rule matches	Usually <i>deny</i> (implicit deny)

Instructor Note:

Emphasize policy hierarchy and rule order evaluation (top-down).

Types of Firewalls

Type	Layer	Description
Packet-Filtering	L3/L4	Filters based on IP, port, protocol
Stateful Inspection	L3-L7	Tracks connection states
Proxy Firewall	L7	Mediates traffic through application proxy
Next-Generation (NGFW)	L7+	Adds DPI, IDS/IPS, and app awareness

3. Packet Inspection

Stateless (Packet-Filtering)

- Examines packets independently
- Rules based on source/destination IP, port, and protocol
- Fast but limited context awareness

Example Rule:

```
ALLOW TCP src=192.168.1.0/24 dst=ANY port=80  
DENY  ALL
```

Weakness:

Cannot detect abnormal sequence or state.

4. Stateful Inspection

Definition:

Tracks active sessions using a *state table*, inspecting packet sequences and connection states.

Advantages:

- Validates TCP handshakes (SYN → SYN/ACK → ACK)
- Blocks unsolicited packets
- More accurate for session-based filtering

Example:

Firewall allows response packets only if they match an existing outbound connection.

Stateful Inspection Flow Example

```
Client (192.168.1.10)
↓ SYN
Firewall (state: new)
↓ SYN/ACK
Server (8.8.8.8)
↓ ACK (state: established)
```

Only packets part of an **established** session are allowed.

Comparing Inspection Methods

Feature	Packet Filter	Stateful Inspection
Speed	Very Fast	Moderate
Context Awareness	None	Tracks sessions
Security Level	Basic	Medium-High
Suitable For	Simple routing/firewalls	Enterprise and cloud gateways

5. Network Address Translation (NAT)

Purpose:

Maps private IP addresses to public ones to conserve address space and conceal internal topology.

Types of NAT:

1. **Static NAT:** One-to-one mapping
2. **Dynamic NAT:** Uses address pool
3. **PAT (Port Address Translation):** Many-to-one mapping

Example (PAT):

```
192.168.1.10:45000 → 203.158.5.2:45000
```

NAT Operation Diagram

```
[Internal Host] 192.168.1.10  
↓  
[Firewall NAT Table]  
↓  
[Public IP] 203.158.5.2  
↓  
[Internet]
```

Benefits:

- Hides internal IP addresses
- Enables Internet access for LAN hosts
- Adds basic layer of obfuscation

NAT Configuration Example (Cisco IOS)

```
Router(config)# interface g0/0
Router(config-if)# ip address 203.158.5.2 255.255.255.0
Router(config)# interface g0/1
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config)# ip nat inside source list 1 interface g0/0 overload
Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255
```

Explanation:

- NAT overload = PAT
- Access-list defines internal IPs eligible for translation

6. Port Forwarding

Definition:

Redirects incoming traffic on a specific port to a designated internal host/service.

Use Cases:

- Remote access to internal web or SSH servers
- Hosting services behind NAT

Example (home router):

```
External: 203.158.5.2:8080  
Forward → 192.168.1.10:80
```

Command Example (iptables):

```
sudo iptables -t nat -A PREROUTING -p tcp --dport 8080 -j DNAT --to 192.168.1.10:80
```

Port Forwarding Diagram

```
[Internet Client]
  ↓ 203.158.5.2:8080
[Firewall / Router]
  ↓ DNAT
[Internal Server 192.168.1.10:80]
```

7. Access Control Lists (ACLs)

Purpose:

Define rules controlling traffic based on IP, protocol, and port.

Type	Direction	Description
Standard ACL	Source IP only	"Permit 192.168.1.0/24"
Extended ACL	Src/Dst + protocol/port	"Permit TCP 192.168.1.0 → Any eq 80"
Named ACL	User-defined label	Easier management

Rule Logic: Top-down evaluation, first match wins, implicit deny at end.

Example: Extended ACL (Cisco)

```
access-list 100 permit tcp 192.168.1.0 0.0.0.255 any eq 80
access-list 100 deny ip any any log
interface g0/0
ip access-group 100 out
```

Meaning:

Allow internal web traffic, deny everything else.

8. Configuration Demonstration (Lab Overview)

Lab Objectives

- Configure basic packet filtering and stateful rules
- Implement NAT and port forwarding
- Apply ACL rules and test connectivity

Tools:

- Cisco Packet Tracer / GNS3
- pfSense / iptables / UFW

Sample pfSense Rule Configuration

Step	Interface	Action	Protocol	Source	Destination	Port
1	LAN	Allow	TCP	192.168.1.0/24	any	80 (HTTP)
2	LAN	Allow	TCP	192.168.1.0/24	any	443 (HTTPS)
3	WAN	Block	All	any	any	*

Result:

Outbound browsing allowed, inbound blocked by default.

Troubleshooting Firewall Issues

Symptom	Possible Cause	Check
No Internet	NAT rule missing	Verify NAT table
Service not reachable	Port forward misconfigured	Inspect DNAT entry
Unexpected access	ACL order incorrect	Review rule priority
Slow traffic	Logging or inspection overhead	Adjust performance settings

9. Firewall Best Practices

- Apply *least privilege* in rulesets
- Maintain **default-deny** policy
- Document rule changes
- Enable **logging and alerts**
- Perform regular audits
- Segment networks with VLANs and subnets
- Combine with IDS/IPS for full visibility

10. Cisco command

Cisco IOS Command Modes

Mode	Prompt	Purpose
User EXEC	Router>	Basic monitoring commands
Privileged EXEC	Router#	Advanced diagnostics, enter config mode
Global Configuration	Router(config)#	System-wide settings
Interface Configuration	Router(config-if)#	Configure specific interfaces
Line Configuration	Router(config-line)#	Manage console, SSH, or VTY access

Cisco IOS Command Modes

Mode Navigation:

```
Router> enable
Router# configure terminal
Router(config)# interface g0/0
Router(config-if)# exit
Router(config)# end
```

Basic Operational Commands

Command	Function
<code>show ip interface brief</code>	Summarize IPs and interface states
<code>show running-config</code>	Display current configuration
<code>show access-lists</code>	Verify ACL rules
<code>show ip nat translations</code>	Check active NAT table
<code>copy running-config startup-config</code>	Save configuration
<code>ping <IP></code>	Test connectivity
<code>traceroute <IP></code>	Verify routing path

##Configuring a Basic Cisco Firewall

Step 1: Define Interfaces and IP Addresses

```
Router(config)# interface g0/0
Router(config-if)# ip address 203.158.5.2 255.255.255.0
Router(config-if)# ip nat outside
Router(config-if)# no shutdown

Router(config)# interface g0/1
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# ip nat inside
Router(config-if)# no shutdown
```

Concept:

- `g0/1` = Inside network (LAN)
- `g0/0` = Outside network (Internet)

Step 2: Configure NAT (Port Address Translation)

```
Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)# ip nat inside source list 1 interface g0/0 overload
```

Verification:

```
show ip nat translations
show ip nat statistics
```

Purpose:

Enables LAN hosts to access the Internet through the router's public IP.

Step 3: Create Basic ACL Rules

Example: Permit HTTP/HTTPS from LAN, Deny Others

```
Router(config)# access-list 100 permit tcp 192.168.1.0 0.0.0.255 any eq 80
Router(config)# access-list 100 permit tcp 192.168.1.0 0.0.0.255 any eq 443
Router(config)# access-list 100 deny ip any any log
```

Apply to Outbound Interface:

```
Router(config)# interface g0/0
Router(config-if)# ip access-group 100 out
```

Step 4: Configure Port Forwarding (Static NAT)

```
Router(config)# ip nat inside source static tcp 192.168.1.10 80 interface g0/0 8080
```

Effect:

External requests to `203.158.5.2:8080` are redirected to internal web server `192.168.1.10:80`.

Step 5: Verify Configuration

Task	Command
Check interface IPs	<code>show ip interface brief</code>
Review ACL hits	<code>show access-lists</code>
Confirm NAT translation	<code>show ip nat translations</code>
Review route table	<code>show ip route</code>
Display logs	<code>show logging</code>

Step 6: Save and Test

```
copy running-config startup-config  
ping 8.8.8.8
```

Expected Result:

- LAN clients reach Internet via NAT.
- External users reach internal server via port forwarding.
- Only permitted traffic passes per ACL rules.

Troubleshooting Cisco Firewall Configuration

Symptom	Cause	Resolution
No Internet access	NAT or ACL missing	Check <code>show ip nat translations</code>
Service unreachable	Incorrect port forward	Verify DNAT rule syntax
Access blocked unexpectedly	ACL order issue	Re-order or insert rules properly
Interface down	Admin-down state	Run <code>no shutdown</code> on interface

Firewall Demonstration: Cisco ACLs & NAT

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Objective:

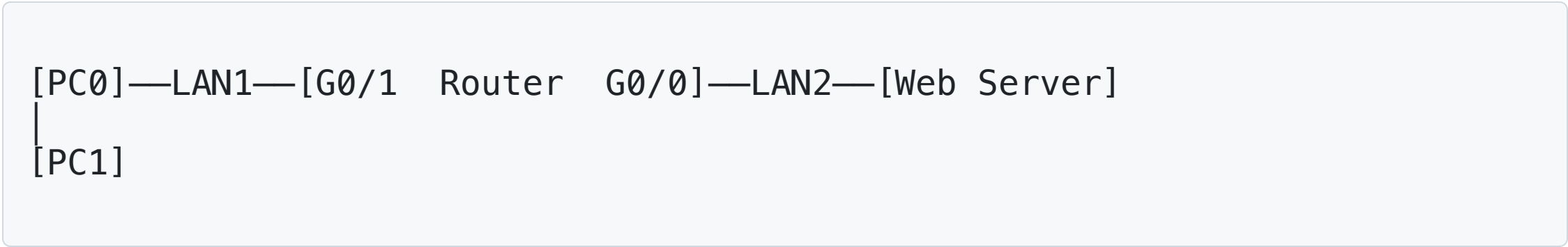
Understand and demonstrate **firewall and ACL configuration** using **Cisco Packet Tracer (v8.2+)**.

Learning Goals

By the end of this session students can:

- Configure basic firewall behavior on a Cisco router
- Implement and test Access Control Lists (ACLs)
- Configure NAT and port forwarding
- Verify and troubleshoot firewall rules

Lab Topology



Device	Interface	IP Address	Role
PC0	NIC	192.168.1.10 /24	Inside LAN
PC1	NIC	192.168.1.11 /24	Inside LAN
Router G0/1	192.168.1.1 /24	Inside	
Router G0/0	203.158.5.1 /24	Outside	
Web Server	NIC	203.158.5.2 /24	External Server

Step 1 – Configure Interfaces

```
enable
configure terminal
!
interface g0/0
 ip address 203.158.5.1 255.255.255.0
 ip nat outside
 no shutdown
exit
!
interface g0/1
 ip address 192.168.1.1 255.255.255.0
 ip nat inside
 no shutdown
exit
```

 Assign IPs to PCs and server in **Desktop** → **IP Configuration**.

Step 2 – Configure NAT (Basic Firewall Behavior)

```
access-list 1 permit 192.168.1.0 0.0.0.255  
ip nat inside source list 1 interface g0/0 overload
```

Verify NAT:

```
show ip nat translations
```

✓ Test from PC0:

ping 203.158.5.2 → should succeed if NAT works.

Step 3 – Apply Access Control List (ACL)

Goal: Allow only HTTP (port 80) to outside world.

```
access-list 100 permit tcp 192.168.1.0 0.0.0.255 any eq 80
access-list 100 deny ip any any log
!
interface g0/0
 ip access-group 100 out
```

✓ Test:

- **Browser (PC0)** → 203.158.5.2 (HTTP) ✓
- **Ping external host** ✗ (should be blocked)

Step 4 – Observe with Simulation Mode

In Packet Tracer:

1. Switch to **Simulation Mode**
2. Edit Filters → select **ICMP, TCP, HTTP**
3. Send a **ping** and a **web request**
4. Observe:
 - HTTP packet passes
 - ICMP blocked at router (ACL deny rule)

Instructor Note:

Pause simulation and highlight rule hits.

Step 5 – Check & Verify

Task	Command
Verify ACL hits	<code>show access-lists</code>
NAT translation	<code>show ip nat translations</code>
Interface status	<code>show ip interface brief</code>
Logs	<code>show logging</code>

Example Output:

```
Deny IP 192.168.1.10 -> 203.158.5.2 (ICMP)
```

Step 6 – Add Port Forwarding (Static NAT)

Expose internal web server:

```
ip nat inside source static tcp 192.168.1.10 80 interface g0/0 8080
```

Test (external client):

`http://203.158.5.1:8080` → forwarded to internal web host.

Step 7 – Troubleshooting Checklist

Symptom	Likely Cause	Command to Check
No Internet	NAT misconfig	<code>show ip nat translations</code>
Ping fails	ACL deny rule	<code>show access-lists</code>
Port forward fails	Wrong inside IP	<code>show running-config</code>
Interface down	Admin down	<code>no shutdown</code>

Step 8 – Instructor Demonstration Ideas

Concept	Activity
ACL priority	Move rule order & re-test
Logging	Enable logging buffered and show denied packets
Security comparison	Contrast router vs pfSense GUI
Student challenge	“Allow HTTPS but deny FTP” task

Cisco Command Reference

Purpose	Command
View interfaces	<code>show ip interface brief</code>
Display config	<code>show running-config</code>
Save config	<code>copy running-config startup-config</code>
Debug packets	<code>debug ip packet detail</code> (then <code>undebug all</code>)
Ping test	<code>ping <IP></code>
Route table	<code>show ip route</code>

Instructor Tips for Packet Tracer v8.2+

- Use **2911** or **1941** routers (support NAT & ACL).
- Enable **HTTP server** service on Web Server node.
- Display CLI Logs window to show ACL matches.
- Zoom into **Simulation Panel** to illustrate packet drops.
- Have students take **screenshots** of successful NAT and ACL hits.

Thank You

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